



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO RKV PETROL BUNK

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

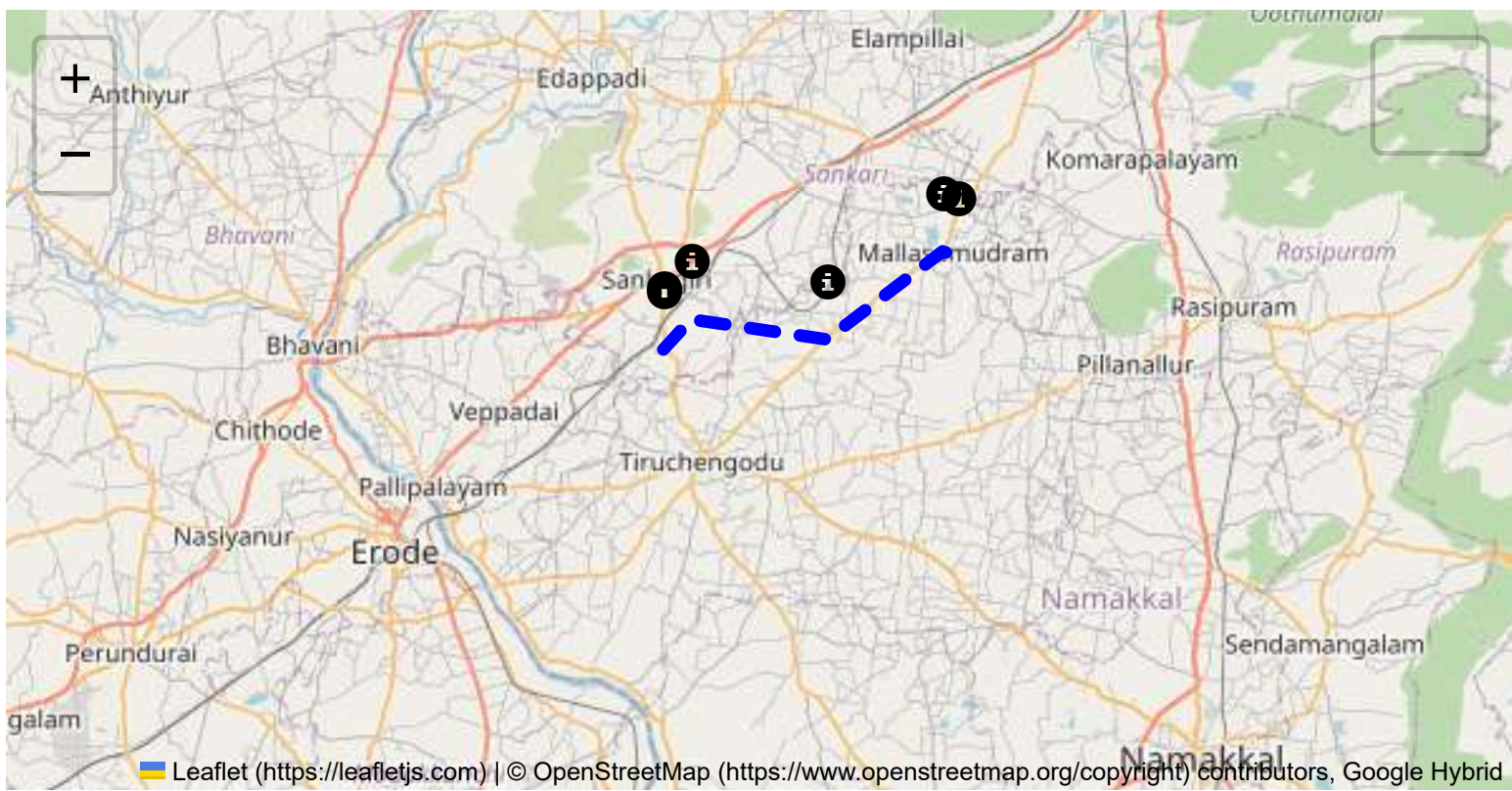
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

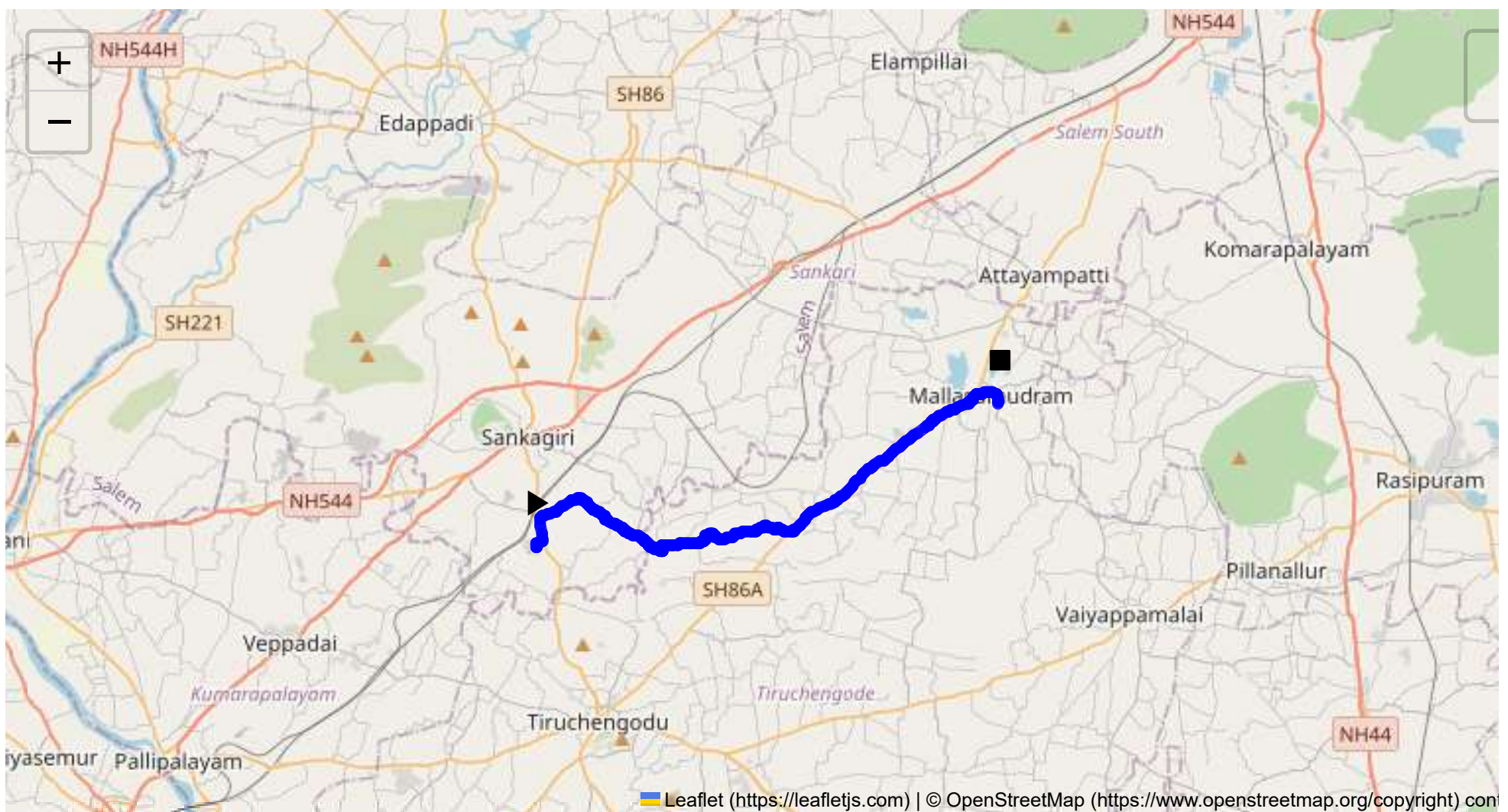
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 22.62 km
Estimated Duration: 0.6 hours
Adjusted Duration (Heavy Vehicle): 0.7 hours
Start: (11.4381, 77.8734)
End: (11.48769, 78.034817)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route stretches approximately 22.62 kilometers from Sangagiri to Mallasamudram in Tamil Nadu, India. It passes through Puthur and incorporates crucial roads like the Sankari R.S Road and Tiruchengode-Salem Main Road. This route mainly comprises local roads and arterial highways traversing semi-urban and rural areas.

2. Typical Weather Conditions and Potential Weather-Related Hazards

Tamil Nadu experiences a tropical climate. The area can witness high temperatures, especially during the summer months (March to May). The monsoon season, from June to September, may cause heavy rainfall, leading to water-logging and slippery road conditions. The period from October to December can also bring some rainfall due to the northeast monsoon. Fog may occasionally occur in early mornings, reducing visibility.

3. Analysis of Traffic Patterns

- **Peak Hours:** Traffic congestion is typically higher during morning (7:00 AM - 9:30 AM) and evening (5:00 PM - 8:30 PM) peak hours.
- **Congestion-Prone Areas:** Puthur and sections near Sangagiri and Tiruchengode often experience slower traffic and bottlenecks due to local commuting and market activities.

4. Assessment of Road Quality and Infrastructure

The road quality varies:

- **Sankari R.S Road and main roads:** Generally well-maintained but may have uneven patches and potholes, particularly post-monsoon.
- **Rural and connector roads:** Subject to wear and tear, narrower, and lack shoulders which could challenge maneuverability, especially for heavy trucks.

5. Suggestions for Alternative Routes for Emergencies

- **Alternative Route:** In emergencies, consider the following path: Take the Tiruchengode - Salem bypass early after Sangagiri, then connect via local roads avoiding major congestion zones.
- **Benefits:** This can help avoid the most congested areas and provide more direct access to the main highways.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Tamil Nadu follows national regulations set by the Ministry of Road Transport and Highways regarding the transport of hazardous materials. Restrictions include:

- Carrying permits for hazardous materials
- Adherence to designated routes and timing restrictions (often at night or early hours).

7. Overview of Historical Incidents

There has been limited reporting on specific historical incidents on these segments. However, typical incidents in the region often involve vehicle breakdowns and minor road accidents due to congestion and road quality.

8. Environmental Considerations and Sensitive Areas

- Agricultural Zones:** The route passes through areas with active farmland; therefore, care should be taken to prevent contamination.
- Habitated Areas:** Proximity to schools and markets in towns like Puthur necessitates cautious driving to avoid accidents.

9. Analysis of Communication Coverage

The route generally has good mobile network coverage, especially near urban areas like Sangagiri and Tiruchengode. Some rural stretches might experience limited connectivity, notably in less populated segments.

10. Estimated Emergency Response Times

- Urban segments (Sangagiri, Tiruchengode):** 15-30 minutes due to proximity to police and health facilities.
- Rural road segments:** Response times might extend to 45-60 minutes due to limited access and availability of emergency services.

11. Overall Summary of Risk Assessment

The route, although moderately risk-prone due to varying road quality and congestion during peak hours, remains manageable with proper precautions. The potential for weather-related disruptions, especially during the monsoon, needs attention. Good communication infrastructure aids in quick emergency coordination, although response times can be improved in rural stretches. For hazardous material transport, adherence to regulatory frameworks and adapting alternate routes when necessary can enhance safety and efficiency.

The driver should be well-versed in local traffic patterns, road conditions, and emergency protocols to navigate this route safely. Regular updates on weather and traffic will be beneficial for optimizing travel plans.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Turn	High	11.43811, 77.87348	15 KM/Hr
1	Turn	Medium	11.43961, 77.87341	30 KM/Hr
2	Turn	Medium	11.43968, 77.87345	30 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
4	Turn	High	11.44861, 77.87429	15 KM/Hr
5	Turn	High	11.43673, 77.91703	15 KM/Hr
6	Turn	High	11.43926, 77.91705	15 KM/Hr
7	Turn	Medium	11.44030, 77.93066	30 KM/Hr
8	Turn	Medium	11.44075, 77.93216	30 KM/Hr
9	Turn	Medium	11.44088, 77.93223	30 KM/Hr
10	Turn	Medium	11.44135, 77.93220	30 KM/Hr
11	Turn	Medium	11.44243, 77.93244	30 KM/Hr
12	Turn	Medium	11.44254, 77.93256	30 KM/Hr
13	Turn	Medium	11.44369, 77.96036	30 KM/Hr
14	Turn	High	11.44387, 77.96283	15 KM/Hr
15	Turn	High	11.49063, 78.02680	15 KM/Hr

Emergency Locations

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	college	Mahendra Engineering College	11.4770412, 78.0024214	30 km/h	Medium
1	school	Akv Schools	11.4850407, 78.0227236	30 km/h	Medium
2	school	Akv Schools	11.4854328, 78.0233558	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44861, 77.87429



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.43673, 77.91703



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.43926, 77.91705



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44030, 77.93066



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44075, 77.93216



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44088, 77.93223



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44135, 77.93220



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44243, 77.93244



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44254, 77.93256



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.44369, 77.96036

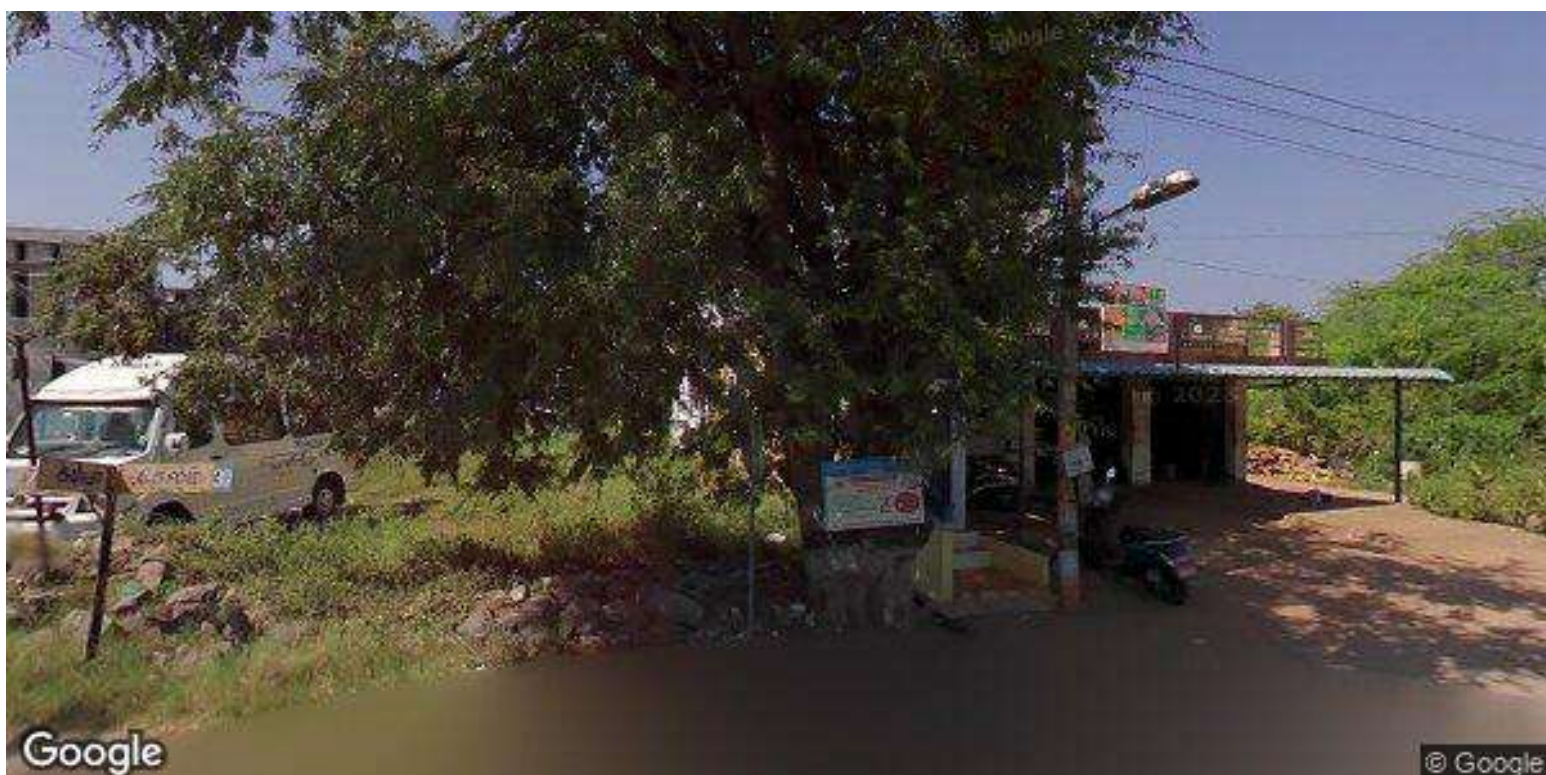


Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44387, 77.96283



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.49063, 78.02680
